

7. (a) Three samples of water, **A**, **B** and **C**, were tested for hardness using soap solution.

The results are shown below.

- No lather formed in samples **A** and **B**
- Lather formed in sample **C**
- When sample **A** was boiled and soap solution added, lather formed
- When sample **B** was boiled and soap solution added, no lather formed

Tick (✓) **three** conclusions that can be drawn from these results.

[3]

sample **C** is soft water

☐

all the samples are hard water

☐

samples **A** and **B** are hard water

☐

sample **B** contains temporary hardness

☐

sample **A** contains temporary hardness

☐

samples **A** and **B** contain permanent hardness

☐

- (b) Give **one** method other than boiling that can be used to remove hardness from water. [1]

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(c) Tick (✓) the compound that causes hardness in water.

[1]

sodium nitrate

☐

zinc chloride

☐

calcium sulfate

☐

potassium oxide

☐

(d) Give **one** health benefit of living in a hard water area.

[1]

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